Faranak Rajabi

Ph.D. Candidate, Mechanical Engineering
M.S., Computer Science (2026)
University of California, Santa Barbara

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Professional Summary

Computational scientist developing high-performance numerical methods for free boundary problems in biological systems, with direct application to **predicting the physical stability of protein-based biotherapeutics** such as monoclonal antibody formulations, and to **optimal control of neural dynamics** for neuromodulation therapies. Author of **five peer-reviewed publications** in computational physics and control systems, with additional work under review. **Invited peer reviewer for 11 manuscripts across four international venues**, including the *Journal of Computational Physics* (Elsevier), the Nature Portfolio, and the IEEE Conference on Decision and Control. Developer of open-source scientific software released to the international research community. Recipient of the **UCSB Graduate Division Dissertation Fellowship (2026–27)**, a campus-wide competitive award, and ranked **184th of 162,879 examinees (top 0.11%)** in the Iranian National University Entrance Examination. Technical expertise spans C++, MPI/CUDA, adaptive mesh refinement, level-set and Hamilton–Jacobi methods, stochastic optimal control, and scientific machine learning.

Education

University of California, Santa Barbara — **Santa Barbara, CA**
Ph.D. in Mechanical Engineering (GPA: 3.94/4.00) — *Jan. 2022 – Expected Jan. 2027*

- **Dissertation (in progress):** *“Free Boundary Problems in Biological Systems: High-Performance Multiscale Simulation of Protein Aggregation and Optimal Control of Neural Dynamics”*
- **Advisors:** Dr. Frédéric Gibou (Editor-in-Chief, *Journal of Computational Physics*) and Dr. Jeff Moehlis (Chair, Department of Mechanical Engineering), Computational Applied Science Laboratory.
- Fully funded throughout doctoral study, with tuition and fees remitted, through competitive fellowships and departmental appointments.
- **Graduate Coursework:** Numerical Simulation with ODEs, Finite Difference Methods for PDEs, Applied Dynamical Systems, Linear Systems Theory, Kalman & Adaptive Filtering, Advanced Matrix Computations, Stochastic Processes.

University of California, Santa Barbara — **Santa Barbara, CA**
M.S. in Computer Science (GPA: 4.00/4.00), Concurrent Second Graduate Degree — *Sep. 2023 – June 2026*

- **Master’s Thesis:** *“Algorithmic Limits in Optimal Control: A Computational and Statistical Study”* (accepted July 2026; ProQuest publication no. 32735436).
- Admitted to a concurrent second graduate degree program with a **full tuition fellowship**, awarded competitively on the basis of academic performance.
- **Core Coursework:** Runtime Systems, Distributed Systems, Data Structures, Advanced Theory of Optimization and Applications, Machine Learning: Signal Processing Approach, Level Set Methods, Group Studies: Cybersecurity and Extended Reality.

Sharif University of Technology — **Tehran, Iran**
B.Sc. in Aerospace Engineering — *Sep. 2016 – Oct. 2021*

- **Admitted with rank 184 nationally** in the Iranian National University Entrance Examination (*Konkur*), Mathematics & Physics track, among **162,879 registered examinees**, placing in the **top 0.11%** nationally. Sharif University of Technology is consistently ranked the top engineering institution in Iran.
- **Full tuition scholarship** for the duration of the degree, awarded on the basis of national entrance examination rank.
- **Undergraduate Thesis:** *“Drug Delivery Applications of Mechanical Micropumps.”* Basis of a subsequent peer-reviewed conference publication.

Research Experience

Graduate Research Assistant **Jan. 2022 – Present**
Computational Applied Science Laboratory, UC Santa Barbara *Santa Barbara, CA*

1. **Continuum Free-Boundary Modeling of Protein Aggregation** *2024–Present*
   - Developing a continuum free-boundary formulation in which monomers form a density field and aggregate surfaces are tracked explicitly, bridging kinetic models, which lack spatial resolution, and particle-based simulation, which is confined to microsecond timescales.
   - Level-set representation of aggregate interfaces on adaptive octree grids, with monomer transport governed by advection-diffusion-reaction dynamics and interface motion set by local mass flux.
   - Designed and validated its numerical foundation: a sharp finite-volume advection-diffusion-reaction solver admitting distinct Robin conditions on segments meeting at kinks, second-order accurate in solution and gradient in L^∞, verified in 2D and 3D on domains with up to eight kinks.
   - *Application:* Aggregation is the principal factor limiting shelf life of high-concentration formulations, a constraint on a U.S. market with 80+ approved monoclonal antibody drugs. The framework forecasts long-term stability from short-term measurements, shortening a cycle now gated by real-time studies.
   - *Implementation:* Parallel C++ within the CASL library, scaling to 1000+ cores; PETSc, MPI, p4est adaptive mesh refinement, level-set methods.
2. **Stochastic Optimal Control for Neural Oscillators** *2022–2026*
   - Developed the first numerical framework for computing globally optimal control in stochastic neural oscillator systems.
   - Event-driven optimal controllers for desynchronizing pathological neural rhythms, relevant to deep brain stimulation for Parkinson's disease and essential tremor.
   - Extended optimal control of coupled oscillators beyond phase reduction, identifying amplitude collapse as a mechanism inaccessible to phase-only models (IEEE CDC 2026).
   - *Technologies:* C++, PETSc, adaptive mesh refinement, numerical optimal control.
3. **Open-Source Scientific Solver Development** *2023–Present*
   - Lead author and principal developer of **CASL-HJX**, an open-source solver for deterministic and stochastic Hamilton–Jacobi equations, released with full documentation and example problems; published in *Computer Physics Communications* (2025). GitHub: UCSB-CASL/CASL-HJX
   - Extending to a GPU-accelerated 4D Hamilton–Jacobi–Bellman solver in CUDA and Kokkos, targeting high-dimensional stochastic control, robotics, and real-time decision systems.
4. **Machine Learning & AI for Science** *2024–2025*
   - Neural signal processing for feature extraction from noisy neurophysiological data; data-driven control combining ML with optimal control theory. *Technologies:* PyTorch, JAX, TensorFlow.

Publications

- **Google Scholar (Aug. 2026):** 24 citations, h-index 3, across 8 publications; **13 from independent research groups** with no co-authorship or institutional connection to the author. First-authored work marked †.
- **Peer-Reviewed Journal Publications:**
  - † **F. Rajabi**, J. Fingerman, A. Wang, J. Moehlis, and F. Gibou, "CASL-HJX: A Comprehensive Guide to Solving Deterministic and Stochastic Hamilton-Jacobi Equations," *Computer Physics Communications*, article no. 109941, 2025. *(CPC archives reusable scientific software for the physics community.)*
  - † **F. Rajabi**, F. Gibou, and J. Moehlis, "Optimal Control for Stochastic Neural Oscillators," *Biological Cybernetics*, vol. 119, no. 2, article no. 9, 2025.
  - M. Zimet, **F. Rajabi**, and J. Moehlis, "Magnitude-Constrained Optimal Chaotic Desynchronization of Neural Populations," *Frontiers in Network Physiology*, vol. 5, article no. 1646391, 2025.
- **Peer-Reviewed Conference Publications:**
  - † **F. Rajabi**, F. Gibou, and J. Moehlis, "Beyond Phase Reduction: Amplitude Collapse in Optimal Control of Coupled Oscillators," *Proc. 65th IEEE Conf. on Decision and Control (CDC)*, Honolulu, HI, December 2026. *(Accepted July 2026; CDC is the IEEE Control Systems Society's flagship conference.)*
  - J. Moehlis, M. Zimet, and **F. Rajabi**, "Nearly Optimal Chaotic Desynchronization of Neural Oscillators," *Proc. 64th IEEE Conf. on Decision and Control (CDC)*, pp. 4797–4802, December 2025.
- **Forthcoming & In Progress:**
  - † **F. Rajabi** and F. Gibou, "Solving Advection-Diffusion-Reaction Equations with Robin Boundary Conditions on Piecewise Smooth Interfaces," under review, 2026. Preprint: SSRN 6802454. *(Already cited.)*

- GPU-accelerated 4D Hamilton–Jacobi–Bellman solver (CUDA/Kokkos), in development; companion release to the IEEE CDC 2026 results.

- **Earlier Peer-Reviewed Publications:**
  - †**F. Rajabi**, A. Bakhshi, and G. Kazemi, "Drug Delivery Applications of Mechanical Micropumps," *Int. Conf. Applied Researches in Science and Eng.*, 2021.
  - Z. Rostami, **F. Rajabi**, and A. Shamloo, "Cell Separation by Using Active and Passive Methods Together," *Proc. 4th Int. Conf. on Innovative Technologies*, 2020.

Invited Service & Independent Recognition

- **Invited Peer Review.** *11 completed review assignments across four international venues, 2025–2026, each undertaken at the direct invitation of a journal editor or conference associate editor.*
  - ***Journal of Computational Physics*** (Elsevier), 2025–Present. **7 assignments**, including second-round reviews of revised manuscripts. *JCP is the field's leading journal for numerical methods in physics.* Topics: Monte Carlo estimators for elliptic interface problems; multigrid methods for ghost finite element approximations; physics-informed neural networks for elliptic interface problems; Fourier neural operators for engineering PDEs.
  - **IEEE Conference on Decision and Control**, 2026. **2 assignments**. Topics: moment closure in stochastic population dynamics; proximal algorithms for composite minimization.
  - ***Communications AI & Computing*** (Nature Portfolio), 2026. **1 assignment**. Topic: surrogate learning of objectives, constraints, and sensitivities for multi-objective optimization of neural dynamical systems.
  - ***Journal of Complex Networks*** (Oxford University Press), 2025. **1 assignment**. Topic: stochastic optimal control for resilience in socio-ecological multiplex networks.

- **Conference Presentations.**
  - **F. Rajabi*** and F. Gibou, "Solving Advection-Diffusion-Reaction Equations with Robin Boundary Conditions on Piecewise Smooth Interfaces," *20th U.S. National Congress on Theoretical and Applied Mechanics (USNCTAM)*, symposium on Scientific Machine Learning and GPU/Exascale Computing for Multiphysics Simulation, Pasadena, CA, June 2026. *(*Presenting author. USNCTAM is the quadrennial national congress of the U.S. theoretical and applied mechanics community.)*
  - **F. Rajabi***, F. Gibou, and J. Moehlis, "Beyond Phase Reduction: Amplitude Collapse in Optimal Control of Coupled Oscillators," *65th IEEE Conference on Decision and Control*, Honolulu, HI, December 2026. *(*Presenting author; scheduled.)*
  - **F. Rajabi***, F. Gibou, and J. Moehlis, "Optimal Control for Stochastic Neural Oscillators," *SIAM Conference on Applications of Dynamical Systems (DS25)*, Denver, CO, May 2025. *(*Presenting author.)*
  - **F. Rajabi***, F. Gibou, and J. Moehlis, "A Level-Set-Method Approach to Optimally Control Stochastic Neural Oscillators," 2023. *(*Presenting author.)*

- **Open-Source Scientific Software.**
  - **CASL-HJX.** Lead author and principal developer of an open-source solver for deterministic and stochastic Hamilton–Jacobi equations, publicly released with full documentation and worked example problems. Published in *Computer Physics Communications*, a journal whose editorial mandate is the dissemination of reusable scientific software to the international research community. github.com/UCSB-CASL/CASL-HJX
  - **GPU-Accelerated 4D Hamilton–Jacobi–Bellman Solver.** CUDA/Kokkos implementation supporting the coupled-oscillator control results in the IEEE CDC 2026 paper, released as a companion repository for reproducibility of the published results.
  - **Supporting solvers.** Poisson solvers on irregular domains, stochastic Hodgkin–Huxley control code, and a PETSc tutorial for parallel scientific computing, all publicly released.

- **Competitively Adjudicated Awards.** The UCSB Graduate Division Dissertation Fellowship (2026–27) is decided by campus-wide faculty review committees in competition across all departments; the Graduate Summer Fellowship was awarded in four consecutive years. Detailed under *Fellowships, Awards & Honors* below.

Fellowships, Awards & Honors

- **UCSB Graduate Division Dissertation Fellowship** (2026–27). Campus-wide central continuing fellowship awarded by faculty review committees across all departments; sole recipient nominated by the Department of Mechanical Engineering.

- **UCSB Graduate Division Summer Travel Grant** (Summer 2026). Competitive funding for international conference travel.

- **UCSB Graduate Summer Fellowship** (2022, 2023, 2024, 2025). Competitive research funding, awarded in four consecutive years.

- **UCSB Conference Travel Grant** (Spring 2025). Competitive funding for research presentation at SIAM DS25.
- **UCSB Computer Science Full Tuition Fellowship** (2023). Full tuition waiver for concurrent second graduate degree, awarded for outstanding academic performance.
- **UCSB Block Grant** (2022). Merit-based award for top incoming doctoral students.
- **Rank 184 of 162,879, Iranian National University Entrance Examination** (2016). National merit examination, Mathematics & Physics track; **top 0.11% nationally**, securing admission to Sharif University of Technology, the highest-ranked engineering institution in Iran.
- **Full Undergraduate Tuition Scholarship**, Sharif University of Technology (2016–2021). Awarded on the basis of national entrance examination rank.

Teaching Experience

Graduate Teaching Assistant — **Spring 2022 – Present**
UC Santa Barbara, Department of Mechanical Engineering — *Santa Barbara, CA*

- Served as TA for **15+ quarter-long appointments** across 6 distinct courses since Spring 2022, instructing over 1000 students in programming, numerical methods, electronics, dynamics, and energy systems.
- Consistently high teaching evaluations: **5.0/5.0** in accessibility and responsiveness to student questions; **4.8/5.0** in clarity of explanations and assignment feedback.
- Created and open-sourced a **comprehensive Modern C++ programming course**: 204 curated code examples across 29 chapters covering C++11/14/17/20, freely available under MIT License on GitHub and used beyond UCSB.

Courses Taught:

- **Introduction to Programming (ME 8):** Fall 2022, Summer 2022, Spring 2023, Summer 2023, Summer 2024, Summer 2025, Fall 2025
- **Mathematics of Engineering (ME 104):** Spring 2022, Fall 2023, Summer 2024, Fall 2024, Summer 2025
- **Introduction to Scientific Computing:** Summer 2025
- **Dynamics:** Summer 2023
- **Basic Electronics & Circuits** (TA & Lab Instructor)**:** Winter 2023
- **Wind & Energy Systems:** Winter 2026

Mentorship, Leadership & Service

- **Invited Peer Reviewer**, four international venues, 2025–Present. Detailed above under *Invited Service & Independent Recognition*.
- **UR2PhD Graduate Mentor**, NSF-funded National Mentorship Program, 2025. Evidence-based mentoring of underrepresented students in computing.
- **APS Career Mentor Fellow**, American Physical Society, 2024–Present. Mentoring 20+ students annually in STEM career development.
- **Undergraduate Research Mentor**, UCSB, 2023–2025. Supervised 2+ students in computational science research; resulted in 1 open-source software package.
- **Women* in Science & Engineering (WiSE) Mentor**, UCSB, 2023–2024.
- **Graduate Hiking & Movement Club President**, UCSB, 2025–Present. Leading a 100+ member interdisciplinary graduate community.

Technical Expertise

- **Numerical Methods & Scientific Computing:** Level-set methods, free boundary and Stefan problems, Hamilton–Jacobi and Hamilton–Jacobi–Bellman equations, deterministic and stochastic PDEs, finite-volume and finite-difference schemes on irregular domains, Poisson and diffusion solvers with Dirichlet/Neumann/Robin conditions, semi-Lagrangian transport, numerical optimal control, Monte Carlo methods.
- **High-Performance Computing:** C++, MPI, CUDA, Kokkos, PETSc, p4est adaptive mesh refinement, parallel algorithm design, performance profiling and optimization, scaling to 1000+ cores on national supercomputing resources.
- **Machine Learning & Data Science:** PyTorch, JAX, TensorFlow, physics-informed and hybrid ML-physics modeling, statistical signal processing, model validation.
- **Software Engineering:** Modern C++ (C++11/14/17/20), Python (NumPy, SciPy), MATLAB, CMake, Git, Linux/Unix, open-source release and documentation practices.